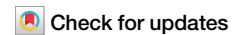


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Deep Sea hydrothermal vent habitats support gravid Antarctic krill



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The long-term resilience of the Southern Ocean ecosystem relies on healthy populations of Antarctic krill, but climate change and increasing fishing pressure are placing this species at risk. Protecting Antarctic krill requires identification of key habitats for critical life history stages. Here, we present multiple lines of evidence that reproductively active female Antarctic krill utilize deep-sea hydrothermal vent habitats. Gravid females collected at Hook Ridge (Bransfield Strait) bore paired spermatophores and exhibited stable isotope signatures, microbial gut communities, and trace-metal profiles consistent with direct consumption of chemosynthetic microbes and prior visitation to vent sites. Complementary ROV footage from Quest Caldera (Scotia Arc) confirms that such vent use is not isolated. Collectively, our findings reveal a previously unrecognized role of vent habitats in the ecology of one of the Southern Ocean's most ecologically critical taxa and highlight the importance of incorporating vent-associated regions into marine protected area planning.

Euphausia superba (hereafter referred to by their common name, Antarctic krill) are vital to the Southern Ocean, supporting vast populations of whales, seals, penguins and other seabirds^{1,2}, and mediating the cycling of essential elements including carbon, nitrogen and iron^{3,4}. Antarctic krill have arguably the highest cumulative biomass of any wild animal species on Earth – nearly half a billion tons⁵. Roughly 30% of their total biomass is concentrated in the region extending from the northern Antarctic Peninsula to the Scotia Sea⁵ (hereafter referred to as nAP-SS for simplicity). This region is one of the fastest warming areas on the planet^{6,7} and is the focus of a growing fishery for Antarctic krill^{8,9}. Consequently, Antarctic krill populations are under increasing pressure, and there is mounting recognition at an international level of the need to conserve critical Antarctic krill habitat to ensure their populations continue to thrive into the future. However, we still lack fundamental knowledge on what constitutes critical habitat for key parts of the Antarctic krill life cycle – particularly for reproductive females¹⁰.

For much of the last century, the conceptual model of Antarctic krill has been that of a primarily epipelagic species, largely occupying the upper ~150–200 m of the water column in the Southern Ocean^{11–14}. In his opus, Marr¹¹ concluded “While it has been established through direct observation... that dense concentrations of the older stages of [krill] ... occur at or very close to the surface of the sea, there is no means of telling ... whether this

congregating of the [species] is a phenomenon exclusive to the surface zone, or whether it be one that may extend into the deeper waters beyond the range of human vision”. It would have been hard to comprehend, in Marr's day, that human vision could extend to the deep seafloor. But advances in underwater video capabilities and deep submersibles in the last half century have enabled new exploration of the deep ocean.

In the austral summer of 1987/88, the first video footage of Antarctic krill near the seafloor was collected on the Weddell Sea Shelf at 416–480 m depth¹⁵. Nearly twenty years later, adult Antarctic krill were seen on video footage from a remotely operated vehicle (ROV) at abyssal depths (up to 3500 m) in Marguerite Trough¹⁶. In January 2010, a benthic video survey off East Antarctica found evidence of krill mating near the seafloor between 400 and 700 m¹⁷. Several other surveys have recorded krill at the seafloor, though typically at depths <1000 m¹⁸. Although those reports are not always considered conclusive – krill may be drawn to and follow the bright lights of a descending ROV, for example – other lines of evidence for Antarctic krill using benthic habitat exist. These include the occurrence of Antarctic krill in the guts of benthic organisms, and the occurrence of benthic prey in the diet of pelagic-caught Antarctic krill, both of which point to the role of krill in the benthic food web¹⁸. However, the role of the deep-sea benthic habitat in the ecology of Antarctic krill remains unclear.

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One of the most significant deep-ocean discoveries of our time has been that of hydrothermal vents at Galápagos Rift¹⁹. Hydrothermal vents support geochemically driven ecosystems sustained by chemolithoautotrophic microbes that oxidize reduced compounds (e.g., sulfide, hydrogen, methane, ferrous iron) to fix inorganic carbon²⁰. Unlike photosynthetic production, which is strongly seasonal in the Southern Ocean²¹, vent-derived chemosynthetic production is comparatively stable and non-seasonal²². Historically framed as isolated “oases” of life in the deep ocean²³, vents are increasingly being recognized as potential sources of energy and biomass for adjacent pelagic communities^{24,25}.

Here, we present multiple lines of evidence that gravid female Antarctic krill utilize active deep-sea hydrothermal vent habitats. In the Antarctic, the primary active vent fields are located in the nAP–SS, including the Bransfield Strait at the northern Antarctic Peninsula and the Scotia Arc^{26,27}. Using a combination of microbiome sequencing, stable isotope analysis, and trace metal signatures made on specimens collected from Hook Ridge (Bransfield Strait), we demonstrate that the ready-to-spawn females had been feeding directly on chemosynthetic microbes. Using video footage obtained during an ROV survey from an expedition to Quest Caldera (Scotia Arc), we show that such behavior is not isolated. Our results reveal a previously unknown aspect of Antarctic krill ecology – that reproductively active females exploit hydrothermal vent ecosystems. We propose that this strategy may confer several important benefits to reproductive females and their offspring.

Results

Survey details

Surveys were conducted at two active hydrothermal vent fields: Hook Ridge (~1100 m depth, Bransfield Strait) and Quest Caldera (~750 m depth, Scotia Arc) using the ROV *SuBastian* aboard two expeditions of the Schmidt Ocean Institute’s research vessel R/V *Falkor (too)* (FKt241214 conducted in December–January 2024 and FKt250220 conducted in February–March 2025). During FKt241214, krill specimens were collected at Hook Ridge and at a comparative, non-vent site in Antarctic Sound (Fig. 1). During FKt250220, an ROV survey was conducted along a grid transect within an area covering approximately 300 × 300 m surrounding Quest Caldera (Fig. 1). The ROV moved at a mean speed of 0.36 m s^{−1} at an altitude of ~30 m above the seafloor. Video footage from the grid survey was analyzed manually for krill by KSB.

Survey site observations

Bottom depths where krill were found ranged between 1020 m and 1160 m at Hook Ridge, between 270 m and 290 m in Antarctic Sound, and between 690 m and 990 m at Quest Caldera. Mean bottom temperatures at each site were −1.21 °C, −1.35 °C, and 0.50 °C, respectively. At Hook Ridge, temperatures within ~2 meters of the vent where Antarctic krill were collected increased to a maximum of 1.38 °C, while temperatures of vent fluid exceeded 200 °C. At Quest Caldera, temperatures within ~2 meters of the vent reached a maximum of 2.52 °C and maximum temperatures recorded in the vent fluid reached 109 °C.

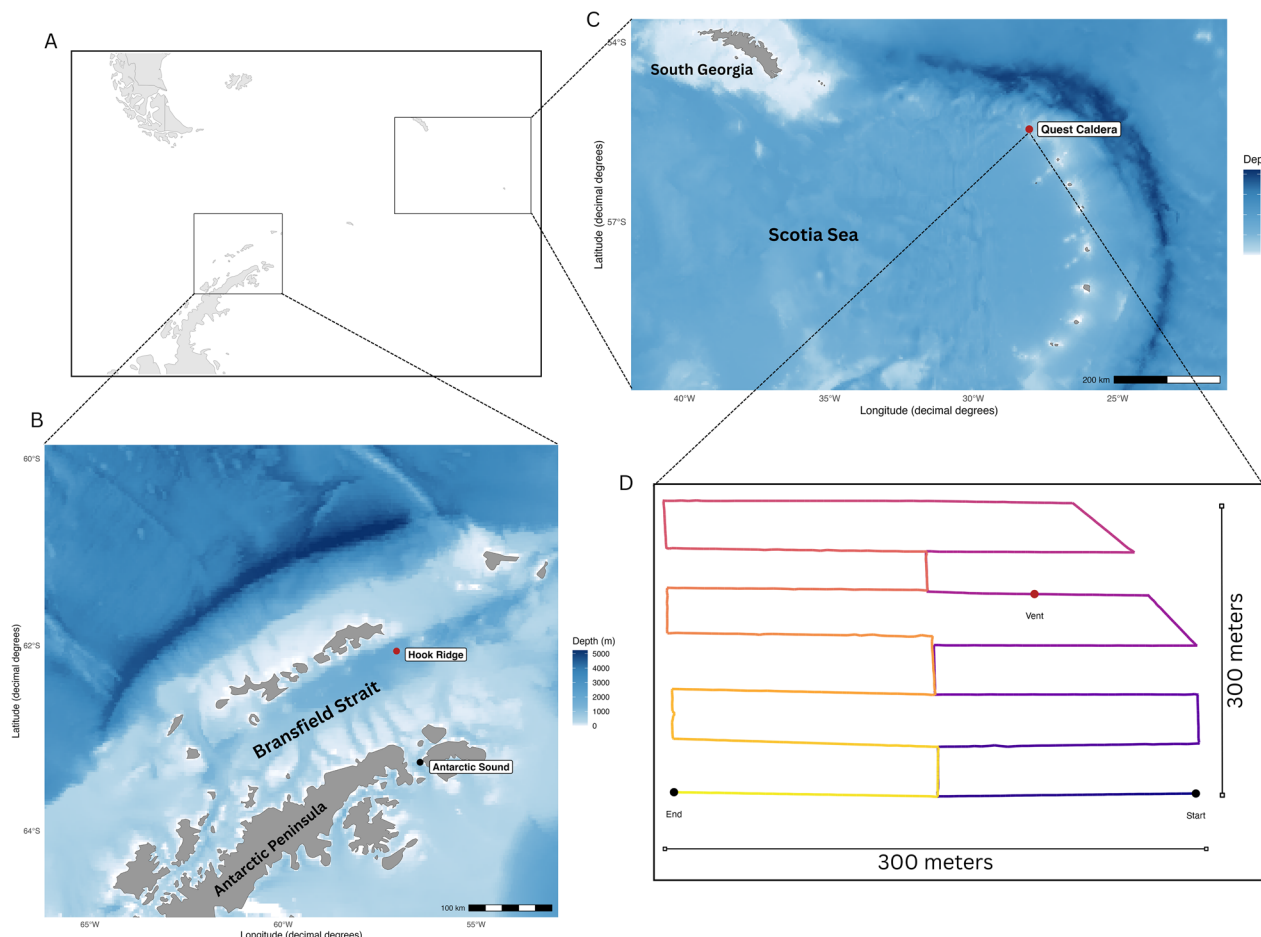
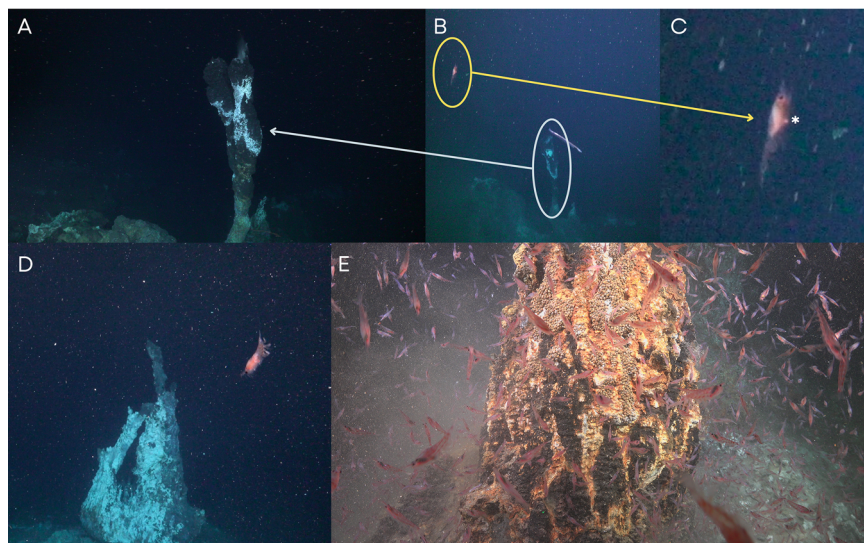


Fig. 1 | Map of study areas. **A** The Atlantic sector of the Southern Ocean, showing the locations of sampling during FKt241214 and FKt250220. **B** Regional bathymetry of the Bransfield Strait and northern Antarctic Peninsula, showing locations of FKt241214 ROV dives at Hook Ridge and Antarctic Sound. **C** Regional bathymetry of the Scotia Sea showing the location of the ROV dive during FKt250220 at Quest

Caldera. South Georgia is shown for spatial context. **D** Schematic of the ROV grid survey conducted at Quest Caldera. The transect lines are colored with a gradient representing time, with survey start in blue and survey end in yellow. The approximate survey footprint of 300 × 300 m is shown.

Fig. 2 | Screenshots taken from ROV dives aboard RV *Falkor (too)*. A–D Screenshots from ROV Dive S0770 at Hook Ridge during FKt241214. **A** A close-up of one of several hydrothermal vent chimneys. **B** A zoomed-out view of the chimney (white oval) in (A) with a gravid female krill (yellow oval). **C** A close-up of the gravid female from (B); note the swollen thoracic region and the spermatophores marked with the asterisk. **D** Another gravid female krill near a different hydrothermal vent chimney at the Hook Ridge site. **E** A dense swarm of gravid female krill around a hydrothermal vent chimney at Quest Caldera during ROV Dive 0804, FKt250220.



Video footage at Hook Ridge showed several gravid female Antarctic krill swimming near the vent structures (Fig. 2A–D). No male krill or other euphausiids were observed at the Hook Ridge vent site during the approximately 9-hour bottom time. Four individuals were collected using the ROV suction sampler and were visibly healthy, displaying normal swimming and escape responses when returned to the ship. In Antarctic Sound, a dense krill swarm was observed at the seafloor at a depth of 288 m, and 35 individuals were collected using the suction sampler. At Quest Caldera, gravid Antarctic krill were observed throughout the ~21-hour bottom time of the Quest Caldera dive, including dense swarms at the active venting site (Fig. 2E).

Demographics of krill from Hook Ridge and Antarctic Sound

The four individuals collected from Hook Ridge were larger than the krill collected from Antarctic Sound. Hook Ridge krill had a mean total length of 47.8 ± 3.9 mm (Table S1). In contrast, the mean total length of Antarctic Sound krill was 31.5 ± 3.0 mm (Table S1). All Hook Ridge krill were sexually mature gravid females (F3D), each with a pair of spermatophores attached to their thelycum. In contrast, krill collected at Antarctic Sound included a mix of juveniles ($n = 10$), males ($n = 16$) and females ($n = 8$; only one advanced sexually to F3D, none held spermatophores) (Table S1).

Gut and vent-associated microbial community composition: Hook Ridge vs. Antarctic Sound

We recovered 754,000 total sequences from the gut microbial and environmental DNA analysis, representing 2527 Amplicon Sequence Variants (ASVs). The gut content was compared to samples of the microbial mats suctioned from the vent edifice and epilithic microbial communities from rocks collected from the hydrothermal vent. The microbial communities from the most active vent-associated substrates (labeled as Rock-White, Rock-Yellow, and Bacterial Mat) were dominated by Campylobacteria, contributing 92–95% to the total microbial composition (Fig. 3A). Within this phylum, the genera *Sulfuromonas* (58–82%) and *Sulfurovum* (4–18%) were most prevalent. *Sulfuromonas* and *Sulfurovum* were also present in the other two substrates (Rock-Black and Rock-Brown), but at lower relative concentrations.

The krill collected from Hook Ridge grouped into two categories: (1) individuals whose gut communities were dominated by vent-associated bacteria, and (2) those with high abundances of the genus *Candidatus Hepatoplasma* (Fig. 3A). Three Hook Ridge krill contained > 40% Campylobacteria in their guts. The fourth individual was dominated (90%) by microbes in the phylum Firmicutes, classified as *Candidatus Hepatoplasma* (or closely related taxa within the same clade). *Candidatus Hepatoplasma*

was also found at lower concentrations in 3 of the 4 rock samples and in all but one krill across both sites.

In contrast, krill from Antarctic Sound had gut communities dominated by Chloroplasts, Actinobacteria, and Rhodobacteriales (Fig. 3A). Chloroplast sequences, which amplify with 16S rRNA primers, are used here as a proxy for ingestion of photosynthetic eukaryotes. In krill from Antarctic Sound, Chloroplasts accounted for 35–51% of gut sequences. Actinobacteriota, primarily the order Corynebacteriales (notably *Rhodococcus* spp.), made up 22–55% of gut communities. Rhodobacteraceae contributed up to 17%. Together, these groups accounted for over 78% of the microbial composition in Antarctic Sound krill.

Gut microbial communities of krill collected at the Hook Ridge vent site were distinct from those of krill collected at Antarctic Sound. Non-metric multidimensional scaling (nMDS) of Bray-Curtis dissimilarities revealed clear separation between gut microbial communities of krill collected at Hook Ridge and Antarctic Sound (nMDS stress = 0.042; Fig. 3B). This pattern was supported by a PERMANOVA, which indicated a significant difference in microbial community composition between sites ($F = 4.03$, $R^2 = 0.25$, $p = 0.001$).

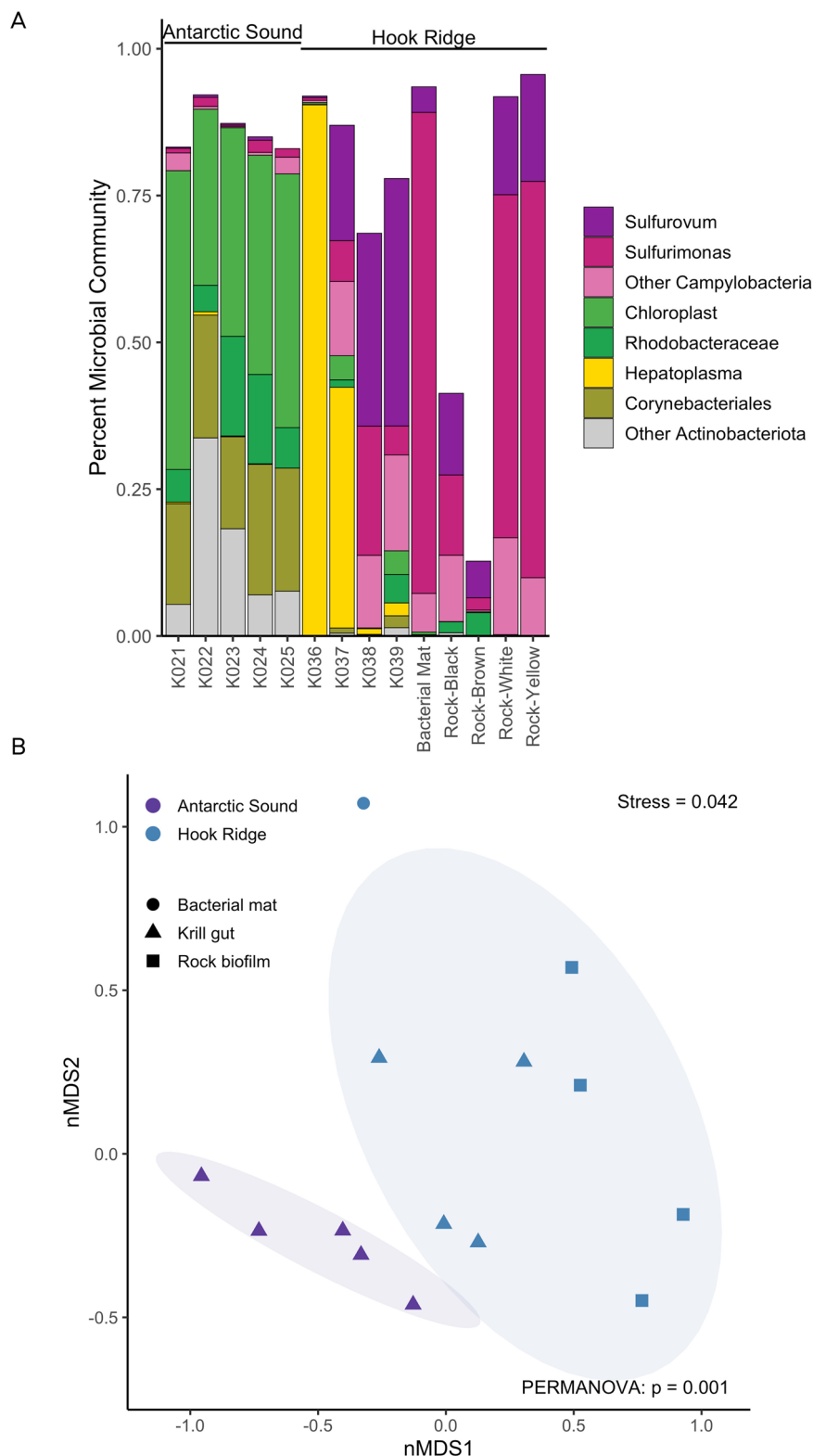
Stable-isotope analysis: Hook Ridge vs. Antarctic Sound

Gravid females from Hook Ridge occupied a distinct cluster in $\delta^{15}\text{N}$ – $\delta^{13}\text{C}$ space (Fig. 4, Table S5), with mean lipid-corrected $\delta^{13}\text{C}$ of $-24.2 \pm 0.5\text{‰}$ and $\delta^{15}\text{N}$ of $3.0 \pm 0.3\text{‰}$, compared to krill collected at Antarctic Sound (mean $\delta^{13}\text{C} = -21.9 \pm 0.6\text{‰}$, $\delta^{15}\text{N} = 3.3 \pm 0.4\text{‰}$). The 95% confidence ellipses showed minimal overlap (Fig. 4), and there was a statistically significant shift in carbon isotope space (ANOVA: $\delta^{13}\text{C}$, $F = 50.84$, $p < 0.001$); though not for $\delta^{15}\text{N}$ (ANOVA: $F = 2.12$, $p = 0.16$).

Trace-metal analysis: Hook Ridge vs. Antarctic Sound

Mean trace metal concentrations of the krill were different between those collected at Hook Ridge and those collected at Antarctic Sound (Table S3). Concentrations of phosphorus-normalized (P-normalized) lead (Pb), zinc (Zn), cadmium (Cd), and barium (Ba) were 2- to 540-fold higher in Hook Ridge krill than those collected at Antarctic Sound (One-Tailed t-Test: $p < 0.05$ in every case; Fig. 5A). P-normalized manganese (Mn) and molybdenum (Mo) concentrations showed modest enrichments (One-Tailed t-Test: $p < 0.05$ for both). Conversely, P-normalized chromium (Cr) and arsenic (As) were lower at Hook Ridge (Fig. 5A) and, because the direction was opposite to the a priori hypothesis, the one-tailed permutation test returned $p = 1.0$ for both. The remaining elements – cobalt (Co), iron (Fe), vanadium (V), copper (Cu), and nickel (Ni) – showed no systematic site effect (One-Tailed t-Test: $p > 0.05$ for all).

Fig. 3 | Microbial community composition of krill guts and hydrothermal-vent substrates. **A** Relative abundance of dominant microbial taxa in krill gut contents and environmental samples from Hook Ridge and Antarctic Sound, based on 16S rRNA amplicon sequencing. Sample types include krill guts, bacterial mats, and rock biofilms. Taxonomic groups are shown at the phylum level (or next highest resolved level). **B** Non-metric multi-dimensional scaling (nMDS) ordination of microbial community composition based on Bray–Curtis dissimilarities calculated from $\log(x + 1)$ -transformed ASV abundances. Symbols distinguish krill gut, bacterial-mat, and rock-biofilm samples, and shaded regions represent 80% normal-distribution ellipses for each sampling location. Differences in microbial community composition between locations were assessed using a one-sided PERMANOVA with 999 permutations ($p = 0.001$). The nMDS stress value is 0.042. The analysis includes $n = 5$ biologically independent samples from Antarctic Sound and $n = 9$ biologically independent samples from Hook Ridge.



Principal component analysis (PCA) of krill trace metal concentrations revealed that the first two principal components explained 93.8% of total variance (Fig. 5B; PC1 = 69.7%, PC2 = 24.1%). PC1 was primarily driven by negative loadings of Ba (−0.40), Zn (−0.40), Cd (−0.40), and Pb (−0.39). Note that although these metals load negatively on PC1, they drive separation toward Hook Ridge samples, which plot on the negative side of PC1, indicating elevated concentrations of hydrothermal metals at that site. PC2 was dominated by high loadings of Cr (0.55), Cu (0.50), and As (0.49). Trace

metal composition differed significantly between krill collected at Hook Ridge and Antarctic Sound (PERMANOVA: $R^2 = 0.797$, $F = 23.56$, $p = 0.025$; Fig. 5B); ~80% of the variation in multivariate metal profiles was attributable to site differences.

Quest Caldera grid survey

During the ~2-hour ROV grid survey at Quest Caldera, gravid female Antarctic krill were observed consistently. Video footage clearly showed that

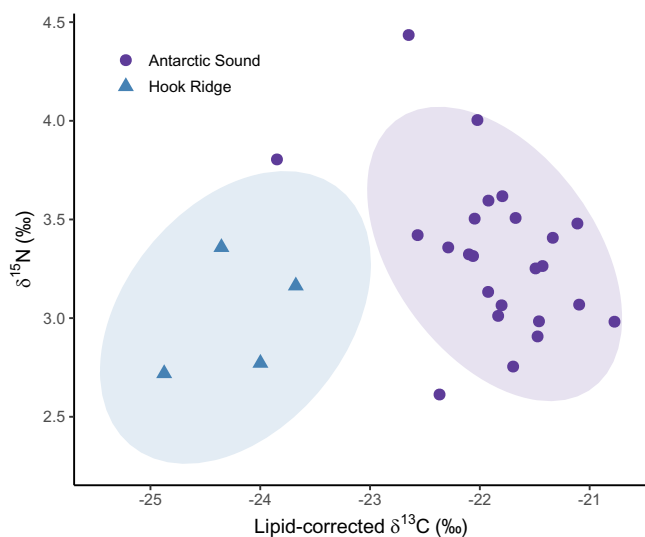


Fig. 4 | Results of stable isotope analysis of krill tissue. Stable isotope values ($\delta^{13}\text{C}$, lipid-corrected, and $\delta^{15}\text{N}$) for Antarctic krill collected from Hook Ridge (blue triangles) and Antarctic Sound (purple circles). Ellipses represent 80% confidence intervals for each site.

many bore spermatophores, indicating readiness to spawn, much like the females collected from Hook Ridge. We estimated there to be 1991 gravid krill along the 2.57 km survey grid, with an average of 0.78 individuals per along-track meter. The integrated number of gravid females estimated per 30-second time-step interval ranged from 0 to 34, with a mean of 8.66 (Fig. 6).

Discussion

Using a combination of analytical approaches conducted on two separate surveys, we provide the first evidence of reproductive Antarctic krill residing and feeding within the chemosynthetic habitats of deep-sea hydrothermal vents.

The gut microbial composition of krill collected at Hook Ridge indicates recent feeding on chemosynthetic microbes. Three of the krill collected at Hook Ridge had been feeding on *Campylobacteria*, including *Sulfurimonas* and *Sulfurovum*. Both are chemosynthetic bacterial genera that are globally present members of hydrothermal vent communities and nearby vent substrates²⁸. In contrast to Hook Ridge, krill collected at Antarctic Sound (non-vent site) had been feeding on phytoplankton and degraded photosynthetic organisms as indicated by the prevalence of Chloroplast sequences in their guts.

The significant difference in $\delta^{13}\text{C}$ between the krill collected at Hook Ridge and Antarctic Sound provides supporting evidence that vent-visiting females had been feeding on a different carbon pool. The lighter $\delta^{13}\text{C}$ values in Hook Ridge krill may be indicative of a chemosynthetic carbon source²⁹. Because $\delta^{13}\text{C}$ in krill reflects an integrated dietary signal accumulated over 3–6 months³⁰, our findings suggest Hook Ridge krill may have fed there intermittently over an extended period. However, the wide range in potential source $\delta^{13}\text{C}$ at vent sites²⁹ and the high degree of variability in tissue-diet shifts in chemosynthetic food webs³¹ make it challenging to definitively attribute specific food sources to the stable isotope signature of Hook Ridge krill.

Diffuse venting at Hook Ridge releases metal-rich, sulfidic fluids that alter the local sediment and porewater geochemistry, leading to the precipitation of polymetallic sulfides, including those rich in Pb, Zn, Ba, and Cd³². These metals are mobilized into the surrounding deep waters and sediments, where they become bioavailable to benthic and pelagic organisms. Concentrations of these metals, and others, were significantly higher in krill collected at Hook Ridge compared to those collected at Antarctic Sound, indicating direct ingestion and/or absorption across gills. Krill

specimen K039, which had the greatest concentration of *Sulfurimonas*, had the highest tissue concentrations of Cu, Mn, Zn, Pb, Ba, Cd, and Fe, supporting a link between feeding on chemosynthetic microbes and metal uptake. Because the trace metal analyses were conducted on krill tissue after guts were removed, the elevated metal concentrations observed in krill from Hook Ridge would have been accumulated in the krill tissue over time and are not a reflection of a singular feeding event, indicating repeated presence near the hydrothermal source.

Together, our results provide compelling evidence that the gravid females at Hook Ridge had been feeding on chemosynthetic material. The ROV survey at Quest Caldera three months later indicates that the association between gravid female krill and hydrothermal vent habitats is not restricted to the Hook Ridge observations. Throughout the ~2-hour grid survey, which represented a subset of the ~21-hour ROV bottom time at Quest Caldera and spanned approximately 300×300 m, gravid females were observed consistently across the area. Based on counts from the ROV footage, we estimated an encounter density of approximately 1 gravid female every ~1.3 meters of ROV track. Because the effective volume of water surveyed by the ROV camera could not be robustly quantified, this estimate cannot be converted directly to a volumetric density. If, purely as an illustrative calculation, the observed encounter rate is treated as equivalent to 0.78 gravid females m^{-3} across the surveyed 300×300 m region, the vent field could contain on the order of 70,000 gravid females.

All specimens observed at the vent and non-vent sites were identified morphologically as *Euphausia superba* (Antarctic krill). Although we did not conduct genetic analyses to test for population structure, we consider it unlikely that the vent specimens represent a distinct species or isolated population. *E. superba* is a highly mobile pelagic euphausiid capable of extensive vertical and horizontal movement, and previous genetic studies have reported weak or absent broad-scale population structure, consistent with substantial connectivity across the Southern Ocean^{33,34}. We therefore interpret the observed differences as reflecting recent habitat use, feeding history, and local environmental conditions rather than population-level divergence or incipient speciation.

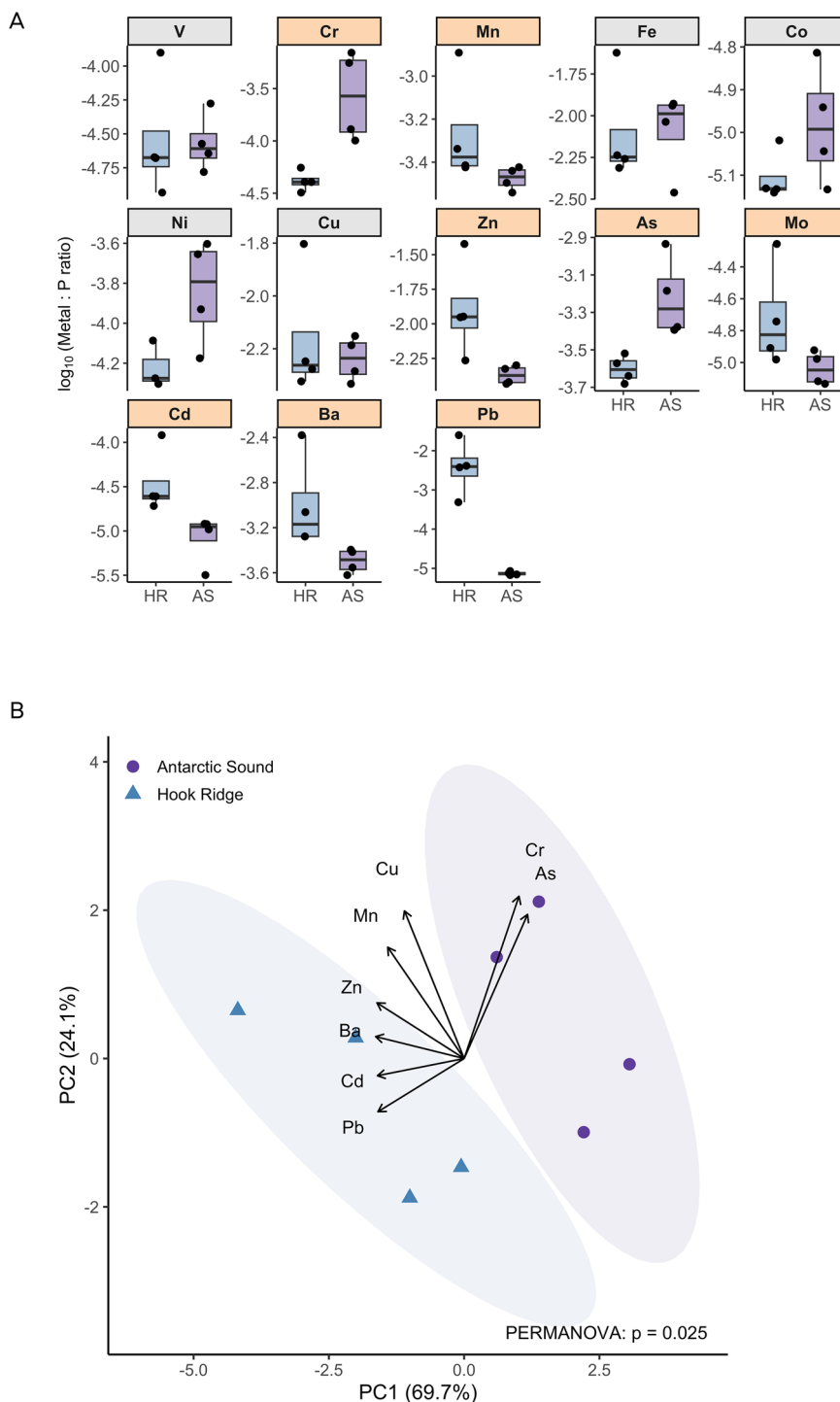
The significance of gravid females at deep-sea hydrothermal vents

The canonical life history model for Antarctic krill asserts that gravid females spawn over deep waters near the continental slope where their embryos have sufficient time to sink through the upper water column and hatch before reaching the seafloor¹⁰. This theoretical model of krill life history underpins our understanding of their biogeography and is used to predict critical spawning ground locations for management purposes¹⁰. Our findings at Hook Ridge, and the complementary observations made at Quest Caldera, reveal a previously unrecognized association between reproductively active female krill and hydrothermal vent habitats that is not currently accounted for in this model.

All four females captured at Hook Ridge (>1000 m depth) bore paired spermatophores, indicating readiness to spawn³⁵, and spermatophores could be clearly seen on gravid females in video footage taken at Quest Caldera. Although the timing between mating and spawning is not explicitly known, it is assumed eggs are released shortly afterwards³⁵. Thus, the presence of spermatophore-bearing gravid females at hydrothermal vent sites between ~700 and >1000 m depth is unexpected under the canonical view that spawning occurs in the upper water column. Whether these females would have released their eggs at depth or ascended to spawn nearer the surface remains unknown. However, either scenario implies that reproductively active females are undertaking deep vertical migrations during the spawning season when their metabolic resources are generally focused on reproductive output. Such extensive migrations are likely to incur substantial energetic costs³⁶, raising the possibility that hydrothermal vents provide physiological, nutritional, or reproductive advantages that outweigh the costs of accessing these habitats.

Seawater temperatures directly surrounding the hydrothermal vent at Hook Ridge were ~1.3 °C, approximately 2.6 °C warmer than seawater away

Fig. 5 | Results of krill tissue trace metal analysis. **A** Phosphate-standardized trace metal concentrations ($\log_{10}$ -transformed) in Antarctic krill collected from Hook Ridge (blue) and Antarctic Sound (purple). Each box plot shows the median, inter-quartile range, and full range for each metal. Each box shows $n = 4$ biologically independent animals from Hook Ridge and $n = 4$ biologically independent animals from Antarctic Sound. Trace metal profiles that differed significantly between sites are highlighted in light orange. **B** Principal components analysis (PCA) of trace metal profiles in Antarctic krill. Individuals from Hook Ridge are shown as blue triangles; individuals from Antarctic Sound are shown as purple circles. Vectors indicate metals with the strongest influence on sample separation. Ellipses represent 80% confidence intervals for each site. PERMANOVA p -value = 0.025.



from the vent. At Quest Caldera seawater temperatures near the vent reached $\sim 2.52^\circ\text{C}$, about 2°C warmer than ambient. At these temperatures, which are within the thermal tolerance of Antarctic krill³⁷ and below the upper thermal threshold (3°C) for krill embryos³⁸, the time required for embryogenesis would be greatly reduced³⁹. Indeed, this strategy of incubating embryos at hydrothermal vents is used by other marine fauna, including the deep-sea skate⁴⁰ and octopus⁴¹. It is also possible that, should spawning occur near the vent site, embryos could be entrained in the upward vertical flow of neutrally-buoyant plume water²⁴. This may provide an effective means of dispersal – a well-known strategy of vent-associated benthic invertebrates⁴² – as vent plumes can extend hundreds of meters above vents and can spread laterally 10 s to 100 s of kilometers²⁴.

Vent sites are also rich in essential trace metals, including Zn and Mn, that are vital to life. Mn concentrations are otherwise extremely low in Southern Ocean waters, the only known ocean region where phytoplankton production is Mn-limited⁴³. Hook Ridge krill had comparatively high levels of Zn and Mn, both of which serve in physiological processes in crustaceans, including reproduction and early development^{44,45}. In the Norway lobster, *Nephrops norvegicus*, Mn load increases in embryos prior to hatching, and is assumed to be associated with development of the gastric mill⁴⁴. Krill embryos bathed in the Mn-rich waters at a hydrothermal vent may therefore benefit from access to this essential, and otherwise sparse, mineral. In addition, there is evidence that both Zn and Mn function in detoxification. For example, Zn may mitigate against the harmful effects of Cd on sperm

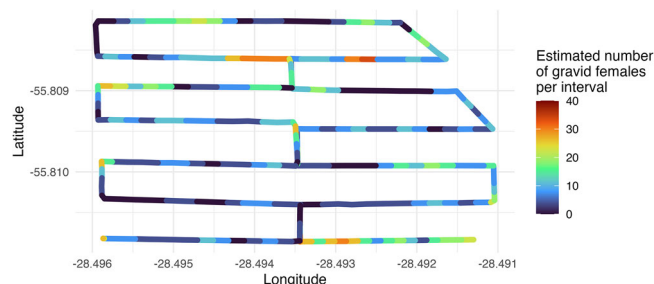


Fig. 6 | Spatial distribution of gravid female Antarctic krill observed during the ROV grid survey at Quest Caldera. The ROV track is shown in geographic space and colored according to the estimated number of gravid females observed during each corresponding 30-s counting interval. Warmer colors correspond to higher estimated abundance. Low-speed or near-stationary intervals were excluded to reduce potential inflation of abundance estimates caused by krill attraction to the ROV.

function in a freshwater crab species⁴⁶, and Mn has been shown to alleviate Cu toxicity in phytoplankton⁴⁷. Thus, the benefits of increased access to essential trace metals may outweigh the potential costs.

Chemosynthetic food webs associated with hydrothermal vents can supply ample food to vent-dwelling and vent-visiting marine fauna⁴⁸, complementing their pelagic phytoplankton-based diet. Female krill that graze on microbial biofilms at hydrothermal vents could acquire phosphatidylethanolamine (PE) and phosphatidylglycerol (PG) – structural components of krill ovarian and embryonic membranes – as these polar lipids are the predominant lipids in *Campylobacteria*^{49,50}. However, the long-chain ω -3 polyunsaturated fatty acids (PUFAs) that females require for successful egg development⁵¹ are not synthesized by *Sulfurimonas* and *Sulfurovum*⁵², and females must obtain these resources from photosynthetic dietary sources. Regardless, the abundance of localized food at hydrothermal vents may be an important subsidy that, unlike photosynthetic production, is available year-round.

Some mutualistic symbionts found in the guts of invertebrates⁵³ can provide a suite of benefits to their hosts, including increasing tolerance to metal toxicity^{54,55}. For instance, the symbiotic relationship between a mercury-resistant bacterium in terrestrial isopods increases the isopod's resilience against mercury toxicity⁵⁴. Earthworms in contaminated environments host Cd-resistant bacteria, benefiting from the reduced toxic loads resulting from the symbiotic relationship⁵⁵. Two of the krill collected at Hook Ridge had *Candidatus Hepatoplasma* in their guts. This less-commonly encountered microbial family is generally considered a symbiont and is known to occur in the hepatopancreas of isopods⁵⁶ and trench amphipods⁵⁷. While its role as a symbiont is still largely unknown, genomic analysis of taxa within this clade suggests it may benefit the host by out-competing potential pathogens⁵⁸, and a symbiont within this genus found in terrestrial isopods has been shown to increase survivorship of their hosts⁵⁹.

Collectively, these thermal, mineral, and nutritional advantages could confer an evolutionary benefit, helping to explain this unexpected behavior.

Implications

Conservation and Management. Antarctic krill are the focus of the largest fishery by tonnage in the Southern Ocean and are central to the structure and functioning of Southern Ocean food webs⁶⁰. Fishing effort is currently concentrated entirely in the nAP-SS⁶¹, where krill also support dense predator populations and where the Domain 1 Marine Protected Area (D1MPA) has been proposed⁶². If vent-associated habitats form part of the broader habitat mosaic used by reproductive female Antarctic krill, as our findings suggest, then hydrothermal vent fields within the Bransfield Strait may warrant explicit consideration in D1MPA planning.

This is particularly relevant because the Bransfield Strait is an actively extending Quaternary back-arc basin³² that contains multiple

submarine volcanic structures aligned along its central rift axis, including Hook Ridge^{63–65}. Although the extent and ecological significance of hydrothermal activity across the basin remain incompletely characterized, our findings indicate that such habitats should not be treated as ecologically isolated from the pelagic krill system. Consequently, it would be prudent for MPA planning in the Domain 1 management area to consider known and potential hydrothermal vent fields within the Bransfield Strait, particularly where these overlap with krill reproductive habitat.

Quest Caldera and other hydrothermal vent fields in the Scotia Arc fall within the South Georgia and South Sandwich Islands MPA, which currently includes protection of the seafloor habitat⁶⁶. Maintaining those protections will be important not only for benthic vent communities, but also for the Antarctic krill population that interacts with the vent-associated environment there. More broadly, hydrothermal vent ecosystems face emerging global pressure from prospective mining of seafloor massive sulfides^{67,68}. Mineral extraction is currently prohibited under the Antarctic Treaty, except for scientific purposes, limiting direct mining pressure in our study region at present⁶⁹. Nevertheless, clarifying the extent of ecological coupling between hydrothermal vents and pelagic food webs remains important for precautionary management, particularly as human interest in deep-sea mineral resources grows⁶⁸.

Conclusions

Overall, our study shows that Antarctic krill ecology may be more closely connected to hydrothermal vents than previously recognized. Our findings not only shed light on a previously undocumented life history strategy for Antarctic krill but broaden the ecological significance of Southern Ocean hydrothermal systems. This work provides new impetus for treating vent-associated regions as potentially important components of the habitat supporting one of the Southern Ocean's most ecologically important species.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The source data for Fig. 3 has been submitted to the NCBI SRA under Bioproject PRJNA1355026. The source data for Fig. 4 are in Supplementary Table S5 and on figshare (<https://doi.org/10.6084/m9.figshare.32978291>). The source data for Fig. 5 are in Supplementary Table S3 and on Figshare (<https://doi.org/10.6084/m9.figshare.32978291>). Krill demographic data are provided in Supplementary Table S1 and on Figshare (<https://doi.org/10.6084/m9.figshare.32978291>). All ancillary data collected during the expedition are available on the Rolling Deck to Repository (<https://doi.org/10.7284/910878>). Video footage and ROV data are archived with KSB, ART, and National Geographic and available upon request.

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Author contributions

Conceptualization: K.S.B.; Investigation: K.S.B., A.F.B., K.N.B., S.C., R.B.F., E.H., R.L.K., A.M., L.E.M., S.S., and A.R.T.; Visualization: K.S.B.; Funding acquisition: K.S.B., K.N.B., A.R.T., A.M.; Project administration: E.H.; Writing – original draft: K.S.B., with specific sections from A.M., L.E.M., and A.R.T.; Writing – review & editing: K.S.B., A.F.B., K.N.B., S.C., R.B.F., E.H., R.L.K., A.M., L.E.M., S.S., and A.R.T.

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Competing interests

The authors declare no competing interests.

Additional information

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